

Trial & Block Structure

CDmem_1.py — Cursor Direction Memory Experiment

■ Timed (fixed duration)

■ Self-paced (waits for participant)

Experiment Overview

Phase	N trials / items	Pacing
GUI dialog	—	Self-paced
QUEST+ Calibration	Min 60 (check: 6) – Max+20; stops at SD < 0.20	Self-paced between phases; timed within trial
Test phase	6 miniblocks × 20 trials = 120 total (check: 6×5 = 30)	Self-paced between phases; timed within trial
Memory test	Test images + foil pairs (variable N)	Self-paced per item

Phase 1 — QUEST+ Calibration

Bayesian adaptive staircase (QUEST+) estimates the motion-influence proportion (s) at which the participant achieves 75% correct shape identification. Stimuli: two shapes (square and circle). **Stopping:** runs min 60 trials (check: 6); stops early if posterior SD < 0.20; hard cap at min+20. A mid-calibration self-paced break is shown at the halfway point.

Per-trial sequence (calibration)

#	Event / Screen	Timing	Duration
1	Algorithm picks optimal proportion s to maximise information gain from the response.	T	< 100 ms
2	White fixation cross on dark background.	T	0.5 – 0.8 s (random uniform)
3	Square and circle placed at $\pm$ offset from centre (sides randomised). Both shapes shown.	S	Waits for any mouse movement
4	BOTH shapes move for exactly 3 s using pre-recorded trajectory snippets. Mouse direction blended into target shape at proportion s . Distractor moves autonomously. No response accepted. EEG triggers sent.	T	3.0 s
5	"Which shape did you control? A = Square S = Circle" shown. Response window opens.	T	5.0 s fixed window
6	"Please answer faster!" if no key pressed in 5 s.	T	2.0 s (only on timeout)
7	"Right" or "Wrong" displayed.	T	0.8 s + 0.3 s blank
8	Posterior updated from response; SD computed.	T	< 100 ms
9	At halfway trial: "Short Break ... Press SPACE to continue."	S	Self-paced — SPACE
10	Steps 1–8 repeat. After min trials: if SD < 0.20 → calibration ends.	—	Min 60 / check 6 trials

Post-calibration (automatic)

#	Event / Screen	Timing	Duration
1	QUEST+ posterior used to find: level_1 = prop at 55% correct (hard); level_3 = prop at 85% correct (medium).	T	Automatic

#	Event / Screen	Timing	Duration
2	quest_threshold_75 and quest_alpha_sd saved to CSV and explInfo.	T	Automatic

Phase 2 — Test Phase

Two difficulty levels: level_1 (55% correct, hardest) and level_3 (85% correct, medium-hard). 6 miniblocks × 20 trials (check: 5). Alternating sequence: Odd participant numbers → L1-L3-L1-L3-L1-L3; Even → L3-L1-L3-L1-L3-L1. Stimuli: two OBJECT IMAGES instead of shapes (test-phase images). Between-miniblock break: 30-second countdown, then SPACE to continue. **No feedback given** (unlike calibration). Agency rating collected instead.

Per-trial sequence (test)

#	Event / Screen	Timing	Duration
1	Shown once before test phase. Explains image task. SPACE to start.	S	Self-paced — SPACE
2	White fixation cross on dark background.	T	0.5 – 0.8 s (random uniform)
3	Two object images placed at ±offset (sides randomised per trial). Participant begins moving mouse.	S	Waits for any mouse movement
4	3-second motion. Mouse direction blended into target image at level_1 or level_3 proportion. EEG triggers at motion start.	T	3.0 s
5	Images reset to start positions. "Which image did you control?" A / S labels shown below images. EEG trigger at screen onset.	T	5.0 s fixed window
6	"Please answer faster!" if no key in 5 s.	T	2.0 s (only on timeout)
7	"How much control did you feel?" 1 (Very weak) – 7 (Very strong). Keyboard 1–7. No feedback.	S	Self-paced
8	Blank screen.	T	0.2 s
9	Steps 2–8 for all 20 trials in miniblock.	—	20 trials / miniblock (check: 5)
10	"Miniblock X of 6 complete! Take a short break. Press SPACE." Then 30-second countdown.	T	30 s countdown + SPACE
11	6 miniblocks total.	—	6 miniblocks

Phase 3 — Memory Test

Old/new recognition for all test images. Shows images that appeared in the test phase plus unseen foils (paired objects not shown). All items randomised. Keys: A = Old (seen during experiment), S = New.

#	Event / Screen	Timing	Duration
1	"The main experiment is now complete. One last part! ... [A = Old, S = New] ... Press SPACE to start."	S	Self-paced — SPACE
2	Single object image shown at screen centre above recognition question.	S	Self-paced — A or S
3	A = Old / S = New key press. Blank screen.	T	0.3 s
4	Steps 2–3 for all memory items.	—	All test + foil images

End of Experiment

#	Event / Screen	Timing	Duration
1	"Thank you for participating! Your data have been recorded. Press SPACE to exit."	S	SPACE to exit
2	CSV (ExperimentHandler), kinematics CSV, and explInfo auto-saved.	T	Automatic